DEPARTMENT OF INDUSTRIAL MANAGEMENT

FACULTY OF SCIENCE | UNIVERSITY OF KELANIYA

Android Fundamentals: UI/UX Flow Design

Wireframing, Prototyping, and Usability Heuristics for Small Screens

Mobile Applications Development (INTE 22283)

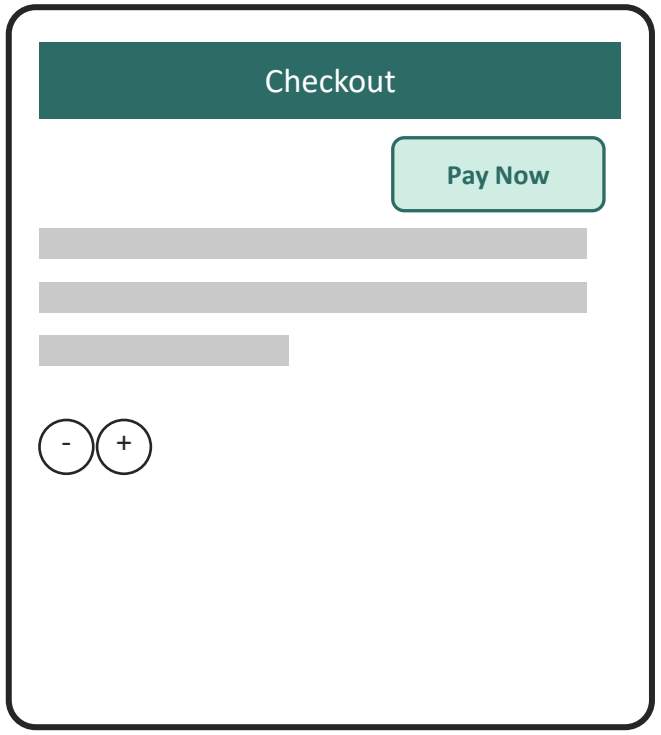
Sammani Maddumahewage · Lecturer (Probationary) · BSc (Hons) in Information Technology

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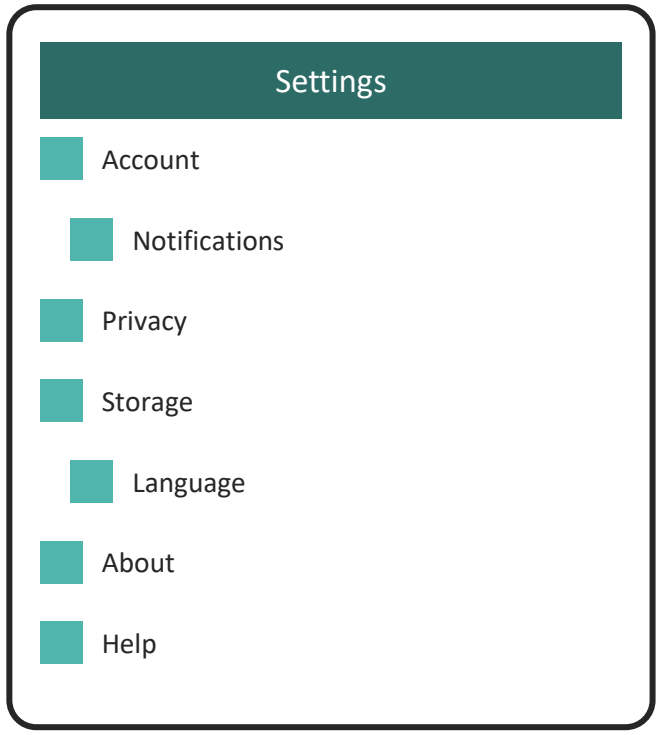
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Practical Activity 1: Interface Issues

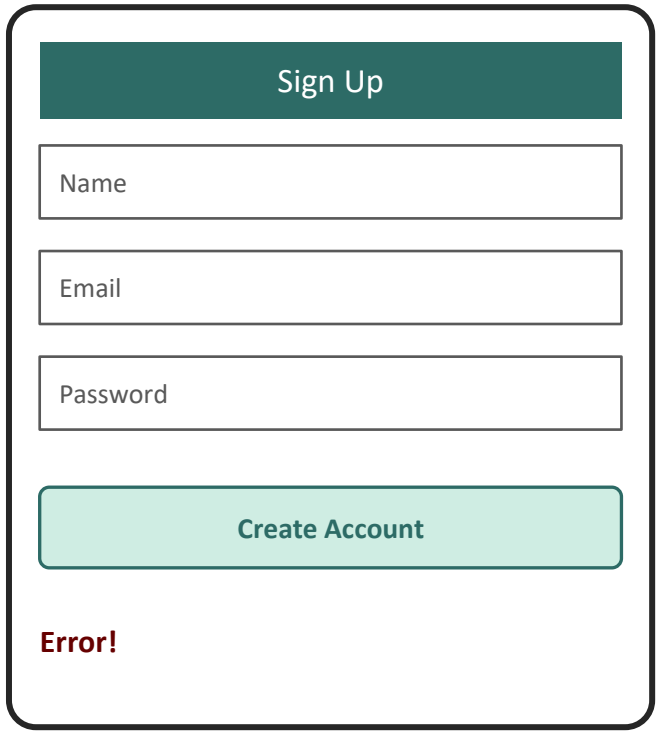
The three screens below belong to QuickCart, a simple shopping app used as a case study throughout this lesson. Examine each screen and identify the usability issues you observe. This activity is completed individually before the discussion. In Practical Activity 2, you will redesign and rebuild these same screens.



Screen A — Checkout



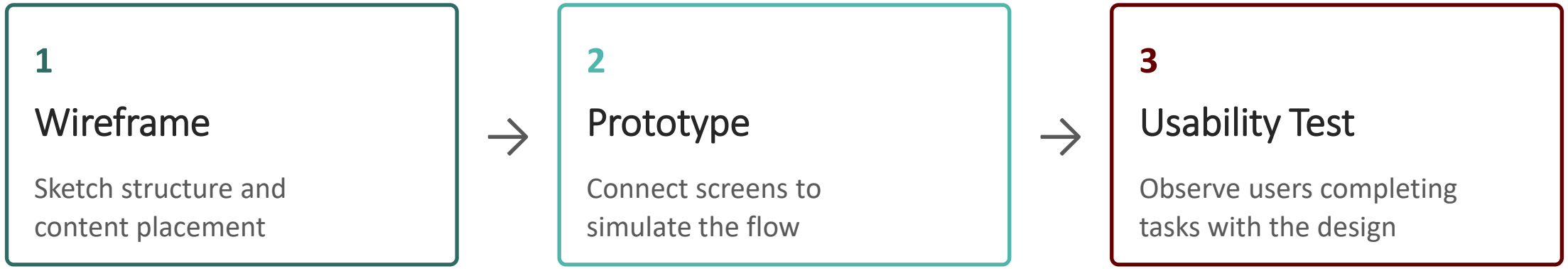
Screen B — Settings



Screen C — Sign Up

The User Interface Design Process

The design of an application flow is carried out in three stages, shown below. Earlier stages are revisited as required based on feedback from later stages.



Identifying a design problem at the wireframe stage is significantly less costly than identifying it after implementation.

Wireframing

A wireframe is a low-fidelity, grayscale layout that shows structure and content placement, without colors, fonts, or icons.

Purpose of wireframing:

- Focuses attention on structure and flow, rather than decoration
- Can be changed quickly, before any code has been written
- Allows the whole team to review and agree on the layout early

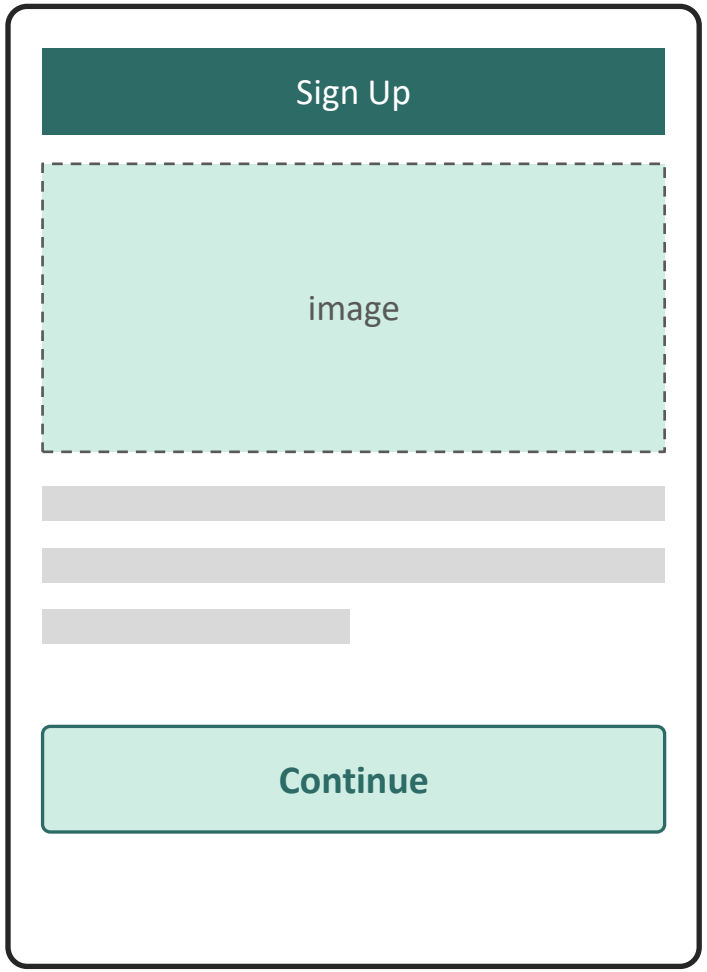
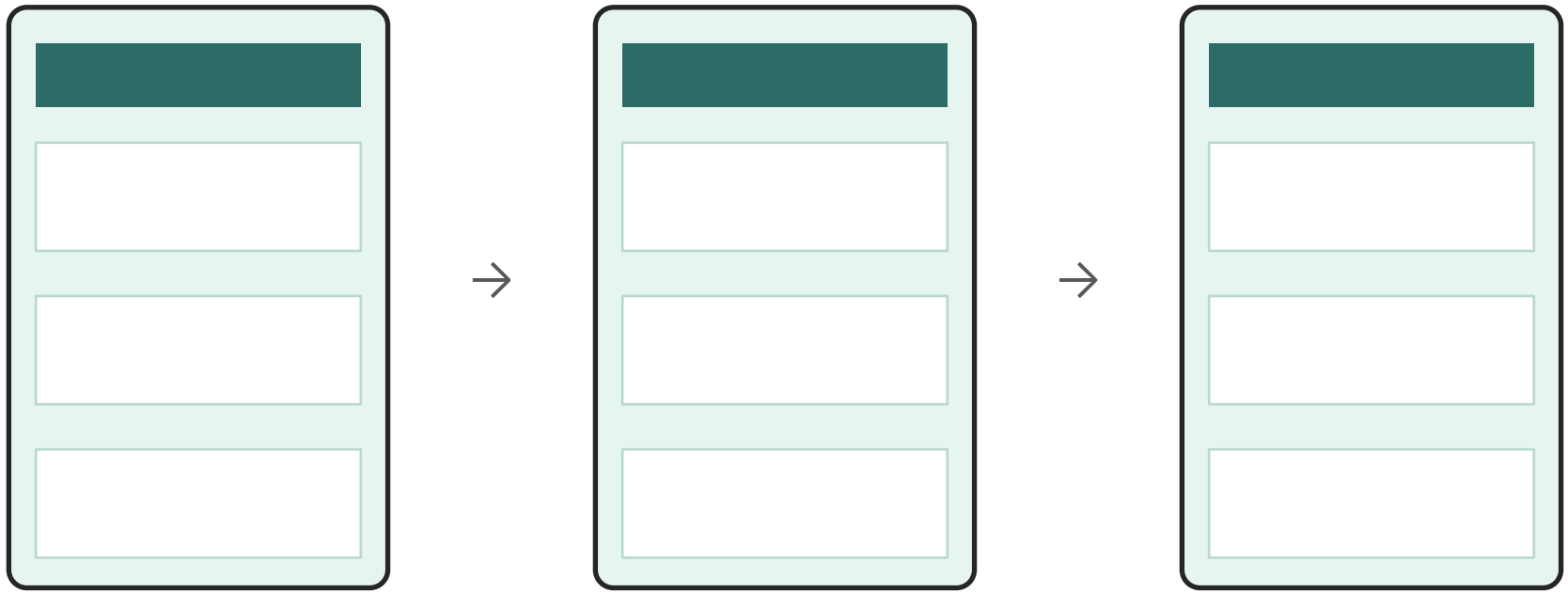


Figure: QuickCart Sign Up wireframe, corrected

Prototyping

A prototype connects a set of wireframes so that the navigation between screens can be reviewed before any application code is written. The example below shows the QuickCart Sign Up, Settings, and Checkout screens connected as a prototype. In this module, prototypes are created using Figma.



Prototype (Figma): the interfaces are linked with clickable hotspots to simulate the flow.
This week's activity: each Figma interface is then embedded in Android Studio as a matching layout and screen.

Layout and Interaction Principles

Support every orientation and size

Verify the flow works in portrait and landscape, on phones and tablets alike

Group related content

Use containment — cards, sections — so structure reads at a glance

Respect safe areas

Keep content clear of system bars, display cutouts, and the on-screen keyboard

Do not overload one screen

Limit the actions per view; split complex tasks across steps

Keep primary actions reachable

Place key interactions within comfortable reach, e.g. a floating action button

Keep spacing and alignment consistent

Uneven gaps make a screen feel disorganized and harder to scan

Source: Android Developers, Design for Android, developer.android.com/design/ui

Usability Heuristics

Adapted from Jakob Nielsen's ten usability heuristics. These principles relate directly to the issues identified in the earlier activity.

Visibility of status

Always show loading or progress state

Consistency and standards

Same button style and placement across screens

Match real-world conventions

Use icons and language users already know

Recognition over recall

Show options rather than expecting users to remember them

User control and freedom

Give a clear back or cancel on every screen

Error prevention

Confirm before destructive actions, such as delete

Design Considerations for Small Screens

Touch targets — Minimum 48dp × 48dp, even if it extends past the icon itself

Legible text — Body text no smaller than 12sp; use sp units so users can scale it

Color contrast — At least 4.5:1 between text and its background

Reach zone — Primary actions low on screen; rare actions placed near the top

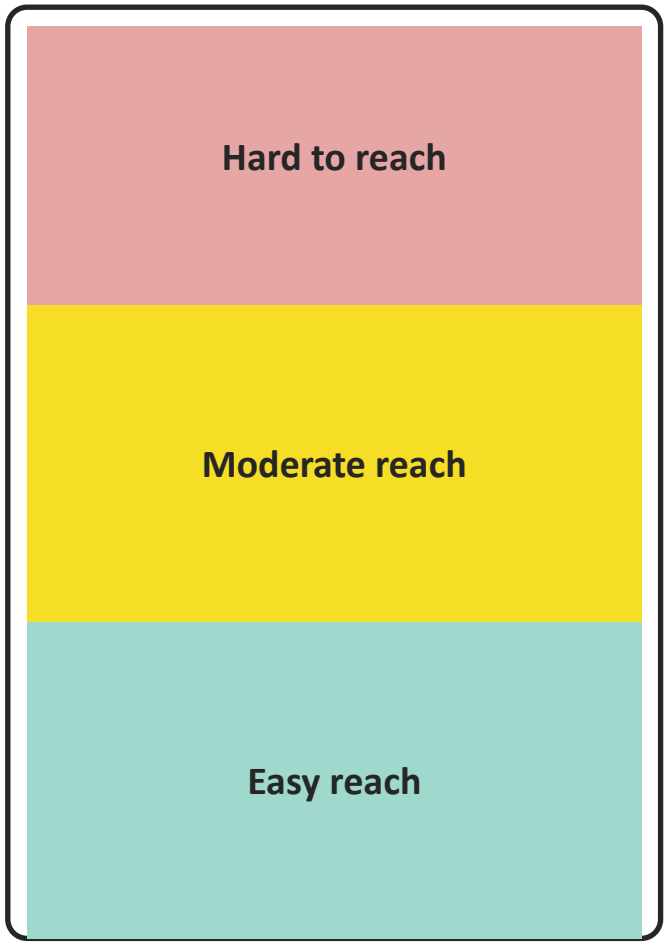


Figure: reach zones on a typical handheld device

Source: Android Developers, Accessibility guidance, developer.android.com/design/ui

Practical Activity 2: Android Build

Case study: QuickCart

Using the Sign Up, Settings, and Checkout screens examined in Practical Activity 1, design corrected interfaces in Figma, then embed each design in Android Studio as the corresponding layout and code.

1. **Design the interfaces in Figma** — Create a corrected Figma design for the Sign Up, Settings, and Checkout screens, applying the principles covered in this lesson

2. **Embed each design in Android Studio** — For every screen designed in Figma, build the matching XML layout and Activity (or Fragment), and write the navigation code between them

3. **Develop a simple working project** — Build and run the project so the Android Studio implementation matches the Figma design for each screen

Deliverables: the Figma file with the corrected QuickCart interfaces, and the Android Studio project (including source code) in which each interface is embedded as a matching layout and screen.

Summary

1. **Wireframe first** — Agree on structure and flow before writing a line of code
2. **Prototype to test the flow** — Not just the visuals — validate that users can move through the app
3. **Heuristics identify problems early** — Cheaper to correct at the wireframe stage than after the app is built
4. **Small screens require careful prioritization** — Show what is essential and place primary actions within reach

Submission Guidelines

Submit the following three items as a single package:

1. **Activity 1 notes** — A short written summary (3–5 points) of the UI/UX issues identified in each of the three QuickCart screens

2. **Figma design file** — The corrected QuickCart interfaces, designed in Figma and exported as a PDF, or shared as a Figma link

3. **Android Studio project** — The project, including source code, in which each Figma interface is embedded as a matching layout and screen

Thank You

Questions

